JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Properties of Triangle

JEE Main Mathematics · Previous Year Questions with Answer Key

34	30	2002 – 2025	29	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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Numerical	3 questions
MCQ (Single Correct Answer)	2 questions
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Answer key	at the end

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.

Numerical 3 questions

Q1. JEE Main 2025 (Online) 22nd January Evening Shift

Let $\mathrm{A}(6,8), \mathrm{B}(10\cos\alpha, -10\sin\alpha)$ and $\mathrm{C}(-10\sin\alpha, 10\cos\alpha)$, be the vertices of a triangle. If $L(a,9)$ and $G(h,k)$ be its orthocenter and centroid respectively, then $(5a-3h+6k+100\sin 2\alpha)$ is equal to _____.

Q2. JEE Main 2024 (Online) 6th April Evening Shift

In a triangle $\mathrm{ABC}, \mathrm{BC}=7, \mathrm{AC}=8, \mathrm{AB}=\alpha\in\mathrm{N}$ and $\cos\mathrm{A}=\frac{2}{3}$. If $49\cos(3\mathrm{C})+42=\frac{\mathrm{m}}{\mathrm{n}}$, where $\gcd(m,n)=1$, then $m+n$ is equal to _____.

Q3. JEE Main 2024 (Online) 4th April Evening Shift

Consider a triangle ABC having the vertices $\mathrm{A}(1,2), \mathrm{B}(\alpha,\beta)$ and $\mathrm{C}(\gamma,\delta)$ and angles $\angle ABC=\frac{\pi}{6}$ and $\angle BAC=\frac{2\pi}{3}$. If the points B and C lie on the line $y=x+4$, then $\alpha^2+\gamma^2$ is equal to _____.

MCQ (Single Correct Answer) 2 questions

Q4. JEE Main 2024 (Online) 9th April Evening Shift

Two vertices of a triangle ABC are $\mathrm{A}(3,-1)$ and $\mathrm{B}(-2,3)$, and its orthocentre is $\mathrm{P}(1,1)$. If the coordinates of the point C are (α,β) and the centre of the of the circle circumscribing the triangle PAB is (h,k), then the value of $(\alpha+\beta)+2(\mathrm{h}+\mathrm{k})$ equals

(A) 81 (B) 15 (C) 51 (D) 5

Q5. JEE Main 2024 (Online) 29th January Morning Shift

Let $\left(5,\frac{a}{4}\right)$ be the circumcenter of a triangle with vertices $\mathrm{A}(a,-2), \mathrm{B}(a,6)$ and $C\left(\frac{a}{4},-2\right)$. Let α denote the circumradius, β denote the area and γ denote the perimeter of the triangle. Then $\alpha+\beta+\gamma$ is

(A) 60 (B) 62 (C) 53 (D) 30

Out of syllabus 29 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q6. JEE Main 2023 (Online) 15th April Morning Shift · **Numerical**

If the line $x = y = z$ intersects the line

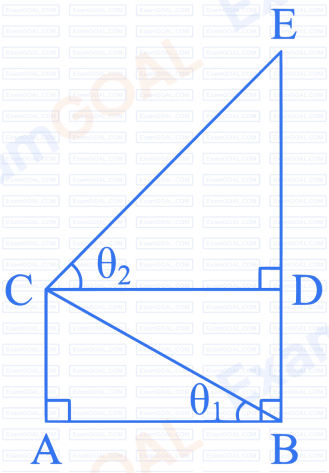
$x \sin A + y \sin B + z \sin C - 18 = 0 = x \sin 2A + y \sin 2B + z \sin 2C - 9$,

where A, B, C are the angles of a triangle ABC, then $80\left(\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}\right)$

is equal to _____.

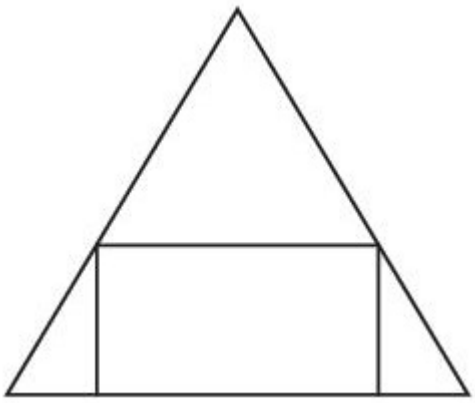
Q7. JEE Main 2023 (Online) 10th April Evening Shift · **Numerical**

In the figure, $\theta_1 + \theta_2 = \frac{\pi}{2}$ and $\sqrt{3}(\mathrm{BE}) = 4(\mathrm{AB})$. If the area of $\triangle \mathrm{CAB}$ is $2\sqrt{3} - 3$ unit ^{2}, when $\frac{\theta_2}{\theta_1}$ is the largest, then the perimeter (in unit) of $\triangle \mathrm{CED}$ is equal to _____.



Q8. JEE Main 2021 (Online) 25th July Evening Shift · **Numerical**

If a rectangle is inscribed in an equilateral triangle of side length $2\sqrt{2}$ as shown in the figure, then the square of the largest area of such a rectangle is _____.



Q9. JEE Main 2021 (Online) 16th March Evening Shift · **Numerical**

In ΔABC, the lengths of sides AC and AB are 12 cm and 5 cm, respectively. If the area of ΔABC is 30 cm^2 and R and r are respectively the radii of circumcircle and incircle of ΔABC, then the value of 2R + r (in cm) is equal to _____.

Q10. JEE Main 2023 (Online) 12th April Morning Shift · **MCQ (Single Correct Answer)**

In a triangle ABC, if $\cos A + 2\cos B + \cos C = 2$ and the lengths of the sides opposite to the angles A and C are 3 and 7 respectively, then $\cos A - \cos C$ is equal to

(A) $\frac{3}{7}$ (B) $\frac{9}{7}$ (C) $\frac{10}{7}$ (D) $\frac{5}{7}$

Q11. JEE Main 2023 (Online) 1st February Morning Shift · **MCQ (Single Correct Answer)**

For a triangle ABC, the value of $\cos 2A + \cos 2B + \cos 2C$ is least. If its inradius is 3 and incentre is M, then which of the following is NOT correct?

(A) $\overrightarrow{MA} . \overrightarrow{MB} = -18$

(B) $\sin 2A + \sin 2B + \sin 2C = \sin A + \sin B + \sin C$

(C) perimeter of ΔABC is $18\sqrt{3}$

(D) area of ΔABC is $\frac{27\sqrt{3}}{2}$

Q12. JEE Main 2022 (Online) 27th June Morning Shift · **MCQ (Single Correct Answer)**

The lengths of the sides of a triangle are $10 + x^2$, $10 + x^2$ and $20 - 2x^2$. If for x = k, the area of the triangle is maximum, then $3k^2$ is equal to :

(A) 5 (B) 8 (C) 10 (D) 12

Q13. JEE Main 2022 (Online) 25th June Morning Shift · **MCQ (Single Correct Answer)**

Let a, b and c be the length of sides of a triangle ABC such that $\frac{a+b}{7} = \frac{b+c}{8} = \frac{c+a}{9}$. If r and R are the radius of incircle and radius of circumcircle of the triangle ABC, respectively, then the value of $\frac{R}{r}$ is equal to :

(A) $\frac{5}{2}$ (B) 2 (C) $\frac{3}{2}$ (D) 1

Q14. JEE Main 2021 (Online) 27th August Morning Shift · **MCQ (Single Correct Answer)**

Let $\frac{\sin A}{\sin B} = \frac{\sin(A-C)}{\sin(C-B)}$, where A, B, C are angles of triangle ABC. If the lengths of the sides opposite these angles are a, b, c respectively, then :

(A) $b^2 - a^2 = a^2 + c^2$ (B) b^2, c^2, a^2 are in A.P.

(C) c^2, a^2, b^2 are in A.P. (D) a^2, b^2, c^2 are in A.P.

Q15. JEE Main 2021 (Online) 20th July Morning Shift · MCQ (Single Correct Answer)

If in a triangle ABC, AB = 5 units, $\angle B = \cos^{-1}\left(\frac{3}{5}\right)$ and radius of circumcircle of ΔABC is 5 units, then the area (in sq. units) of ΔABC is :

(A) $10 + 6\sqrt{2}$ (B) $8 + 2\sqrt{2}$ (C) $6 + 8\sqrt{3}$ (D) $4 + 2\sqrt{3}$

Q16. JEE Main 2021 (Online) 26th February Evening Shift · MCQ (Single Correct Answer)

The triangle of maximum area that can be inscribed in a given circle of radius 'r' is :

(A) An equilateral triangle having each of its side of length $\sqrt{3}$r.

(B) An equilateral triangle of height $\frac{2r}{3}$.

(C) A right angle triangle having two of its sides of length 2r and r.

(D) An isosceles triangle with base equal to 2r.

Q17. JEE Main 2021 (Online) 24th February Evening Shift · MCQ (Single Correct Answer)

Let a, b, c be in arithmetic progression. Let the centroid of the triangle with vertices (a, c), (2, b) and (a, b) be $\left(\frac{10}{3}, \frac{7}{3}\right)$. If α, β are the roots of the equation $ax^2 + bx + 1 = 0$, then the value of $\alpha^2 + \beta^2 - \alpha\beta$ is :

(A) $\frac{69}{256}$ (B) $\frac{71}{256}$ (C) $-\frac{71}{256}$ (D) $-\frac{69}{256}$

Q18. JEE Main 2020 (Online) 4th September Morning Slot · MCQ (Single Correct Answer)

A triangle ABC lying in the first quadrant has two vertices as A(1, 2) and B(3, 1). If $\angle BAC = 90^o$ and area$(\Delta ABC) = 5\sqrt{5}$ s units, then the abscissa of the vertex C is :

(A) $1 + 2\sqrt{5}$ (B) $2\sqrt{5} - 1$ (C) $1 + \sqrt{5}$ (D) $2 + \sqrt{5}$

Q19. JEE Main 2019 (Online) 10th April Evening Slot · MCQ (Single Correct Answer)

The angles A, B and C of a triangle ABC are in A.P. and a : b = 1 : $\sqrt{3}$. If c = 4 cm, then the area (in sq. cm) of this triangle is :

(A) $2\sqrt{3}$ (B) $4\sqrt{3}$ (C) $\frac{4}{\sqrt{3}}$ (D) $\frac{2}{\sqrt{3}}$

Q20. JEE Main 2019 (Online) 8th April Evening Slot · MCQ (Single Correct Answer)

If the lengths of the sides of a triangle are in A.P. and the greatest angle is double the smallest, then a ratio of lengths of the sides of this triangle is :

(A) 5 : 9 : 13 (B) 5 : 6 : 7 (C) 4 : 5 : 6 (D) 3 : 4 : 5

Q21. JEE Main 2019 (Online) 11th January Evening Slot · MCQ (Single Correct Answer)

Given $\frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13}$ for a ΔABC with usual notation.

If $\frac{\cos A}{\alpha} = \frac{\cos B}{\beta} = \frac{\cos C}{\gamma}$, then the ordered triad (α, β, γ) has a value :

(A) (19, 7, 25) (B) (7, 19, 25) (C) (5, 12, 13) (D) (3, 4, 5)

Q22. JEE Main 2019 (Online) 11th January Morning Slot · MCQ (Single Correct Answer)

In a triangle, the sum of lengths of two sides is x and the product of the lengths of the same two sides is y. If $x^2 - c^2 = y$, where c is the length of the third side of the triangle, then the circumradius of the triangle is :

(A) $\frac{y}{\sqrt{3}}$ (B) $\frac{c}{3}$ (C) $\frac{c}{\sqrt{3}}$ (D) $\frac{3}{2}$y

Q23. JEE Main 2019 (Online) 10th January Evening Slot · MCQ (Single Correct Answer)

With the usual notation, in ΔABC, if $\angle A + \angle B = 120^o$, a = $\sqrt{3}+1$, b = $\sqrt{3}-1$ then the ratio $\angle A : \angle B$, is :

(A) 9 : 7 (B) 7 : 1 (C) 5 : 3 (D) 3 : 1

Q24. JEE Main 2018 (Offline) · MCQ (Single Correct Answer)

Let the orthocentre and centroid of a triangle be A(-3, 5) and B(3, 3) respectively. If C is the circumcentre of this triangle, then the radius of the circle having line segment AC as diameter, is :

(A) $\frac{3\sqrt{5}}{2}$ (B) $\sqrt{10}$ (C) $2\sqrt{10}$ (D) $3\sqrt{\frac{5}{2}}$

Q25. AIEEE 2012 · MCQ (Single Correct Answer)

In a ΔPQR, If $3\sin P + 4\cos Q = 6$ and $4\sin Q + 3\cos P = 1$, then the angle R is equal to :

(A) $\frac{5\pi}{6}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{4}$ (D) $\frac{3\pi}{4}$

Q26. AIEEE 2010 · MCQ (Single Correct Answer)

For a regular polygon, let r and R be the radii of the inscribed and the circumscribed circles. A $false$ statement among the following is :

(A) There is a regular polygon with $\frac{r}{R} = \frac{1}{\sqrt{2}}$

(B) There is a regular polygon with $\frac{r}{R} = \frac{2}{3}$

(C) There is a regular polygon with $\frac{r}{R} = \frac{\sqrt{3}}{2}$

(D) There is a regular polygon with $\frac{r}{R} = \frac{1}{2}$

Q27. AIEEE 2005 · MCQ (Single Correct Answer)

In a triangle ABC, let $\angle C = \frac{\pi}{2}$. If r is the inradius and R is the circumradius of the triangle ABC, then $2(r+R)$ equals :

(A) $b+c$ (B) $a+b$ (C) $a+b+c$ (D) $c+a$

Q28. AIEEE 2005 · MCQ (Single Correct Answer)

If in a ΔABC, the altitudes from the vertices A, B, C on opposite sides are in H.P, then $\sin A, \sin B, \sin C$ are in :

(A) G. P. (B) A. P. (C) A.P-G.P. (D) H. P

Q29. AIEEE 2004 · MCQ (Single Correct Answer)

The sides of a triangle are $\sin\alpha$, $\cos\alpha$ and $\sqrt{1+\sin\alpha\cos\alpha}$ for some $0 < \alpha < \frac{\pi}{2}$. Then the greatest angle of the triangle is :

(A) 150° (B) 90° (C) 120° (D) 60°

Q30. AIEEE 2003 · MCQ (Single Correct Answer)

The sum of the radii of inscribed and circumscribed circles for an n sided regular polygon of side a, is :

(A) $\frac{a}{4}\cot\left(\frac{\pi}{2n}\right)$ (B) $a\cot\left(\frac{\pi}{n}\right)$

(C) $\frac{a}{2}\cot\left(\frac{\pi}{2n}\right)$ (D) $a\cot\left(\frac{\pi}{2n}\right)$

Q31. AIEEE 2003 · MCQ (Single Correct Answer)

In a triangle ABC, medians AD and BE are drawn. If $AD = 4$, $\angle DAB = \frac{\pi}{6}$ and $\angle ABE = \frac{\pi}{3}$, then the area of the $\angle\Delta ABC$ is :

(A) $\frac{64}{3}$ (B) $\frac{8}{3}$ (C) $\frac{16}{3}$ (D) $\frac{32}{3\sqrt{3}}$

Q32. AIEEE 2003 · MCQ (Single Correct Answer)

If in a ΔABC $a\cos^2\left(\frac{C}{2}\right) + c\cos^2\left(\frac{A}{2}\right) = \frac{3b}{2}$, then the sides a, b and c :

(A) satisfy $a+b=c$ (B) are in A.P (C) are in G.P (D) are in H.P

Q33. AIEEE 2002 · MCQ (Single Correct Answer)

The sides of a triangle are $3x+4y$, $4x+3y$ and $5x+5y$ where $x, y > 0$ then the triangle is :

(A) right angled (B) obtuse angled (C) equilateral (D) none of these

Q34. AIEEE 2002 · MCQ (Single Correct Answer)

In a triangle with sides a, b, c, $r_1 > r_2 > r_3$ (which are the ex-radii) then :

(A) $a > b > c$ (B) $a < b < c$ (C) $a > b$ and $b < c$ (D) $a < b$ and $b > c$

Answer Key

Numerical

1. **145**	2. **39**	3. **14**

MCQ (Single Correct Answer)

4. **D**	5. **C**

Out of syllabus

6. **5**	7. **6**	8. **3**	9. **15**	10. **C**
11. **D**	12. **C**	13. **A**	14. **B**	15. **C**
16. **A**	17. **C**	18. **A**	19. **A**	20. **C**
21. **B**	22. **C**	23. **B**	24. **D**	25. **B**
26. **B**	27. **B**	28. **B**	29. **C**	30. **C**
31. **D**	32. **B**	33. **B**	34. **A**	